What Makes a Board Game Great? Mining BoardGameGeek

mini-project technical report

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A note on contributions and the use of AI. Our differing levels of familiarity with Git and GitHub resulted in uneven commit counts in the repository history; commit counts do not reflect effort, and all three team members contributed equally across data collection, preprocessing, exploratory analysis, machine learning, and the web application. We used AI-assisted tools to help with code writing, debugging, and polishing this report; every AI-generated output was reviewed and validated by the team, who take full responsibility for the final results. More details can be found in the project README.

Abstract

The board game market has grown rapidly, yet the quality of newly released games is hard to judge before player ratings accumulate. We confront this **cold-start problem** using data from BoardGameGeek (BGG): a complete pipeline collects 2,077 base games via BGG's offcial XML API2, and predicts two complementary quantities–*popularity* (regression, $R^2 \approx 0.73$) and *quality* (binary classification, test AUC up to 0.856) – from a deliberately rating-free feature set of 10 structured attributes plus 30 latent text components from game description text themes as the strongest quality signal (71% of Gini importance; shuffling the text block drops test AUC by 0.078), and reveal that description length is a redundant proxy. After tuning the decision threshold on the precision–recall curve, we get an F1 of 0.71, compared with a 69.2% majority-class baseline. This is comparable to a leakage-prone variant that used popularity as a feature. We then turn the predictions into an interpretable content-based gamer recommender/finder and an interactive Streamlit app.

Keywords: cold-start problem, recommender systems, text mining, board games, interpretability.

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